

Build an AI Talent Acquisition Platform

Build a production-ready recruitment platform inspired by modern HR systems such as KayanHR's Talent Acquisition and Employee Self-Service modules, but with significantly stronger AI capabilities. The platform should cover recruitment, candidate management, AI-assisted hiring, analytics, and workflow automation.

Think of this as:

"If DigitalSofts were building the next-generation AI recruitment platform."

Duration

7–10 Days

Objective

Design and build a SaaS application that could realistically be sold to HR departments.

The solution should demonstrate:

- AI Engineering
- Backend Engineering
- Frontend Engineering
- System Design
- Production Readiness
- Security
- DevOps
- Scalability

Tech Stack

Backend

- FastAPI
- PostgreSQL
- SQLAlchemy
- Alembic

- Redis
- Celery / Dramatiq / Background Workers

Frontend

- Next.js 15
- React
- Tailwind
- TypeScript

Authentication

- JWT
- OAuth (Google)

Deployment

- Docker
- Docker Compose

Optional

- Kubernetes
- GitHub Actions

Core Modules

1. Authentication

Implement

- Register
- Login
- Refresh Tokens
- Password Reset
- Email Verification
- Google Login

Roles

- Super Admin
- HR Manager
- Recruiter
- Interviewer

2. Organization Management

Multiple organizations

DigitalSofts
ABC Software
XYZ Bank

Each organization owns

- users
- candidates
- jobs
- interviews
- reports

Implement proper tenant isolation.

3. Recruitment Module

Create

- Departments
- Jobs
- Hiring Pipeline

Example pipeline

Applied

↓

Screening

↓

Interview

↓

Technical

↓

HR



Offer



Hired

Drag-and-drop preferred.

4. Candidate Portal

Candidate can

- Register
- Upload Resume
- Track Status
- Update Profile
- Receive Notifications

5. Resume Parsing

Extract automatically

- Name
- Email
- Phone
- Skills
- Experience
- Education
- Certifications
- Projects

Store structured data.

6. AI Candidate Matching

Given

Resume

-

Job Description

Generate

- Match %
- Missing Skills
- Strengths
- Weaknesses
- Hiring Recommendation
- Interview Questions

Do NOT simply send the resume to ChatGPT.

Use

- Embeddings
- Hybrid Search
- RAG
- Metadata filtering

7. AI Recruiter Copilot

Allow HR to ask

Find Python developers with Kubernetes experience

Show candidates who worked on RAG

Who has cloud certifications?

Natural language should become database queries.

8. Interview Scheduler

Calendar

Availability

Email invitation

Meeting Link

9. AI Interview Feedback

After interview notes

Generate

- Summary
- Strengths
- Weaknesses
- Recommendation
- Overall Score

10. Analytics Dashboard

Charts

- Hiring Funnel
- Time to Hire
- Offer Acceptance
- Candidate Sources
- Recruiter Performance

Production Requirements

API

Swagger

OpenAPI

Versioning

/api/v1/

Architecture

Use proper layers

Controllers

↓

Services

↓

Repositories

↓

Database

No business logic inside routes.

Security

Implement

- Rate limiting
- RBAC
- Input validation
- SQL Injection prevention
- XSS prevention
- Secure Headers
- File validation

Logging

Structured logs

Request IDs

Error IDs

Error Handling

Consistent JSON

Example

```
{  
  "success": false,  
  "error": {  
    "code": "JOB_NOT_FOUND",  
    "message": "Job does not exist"  
  }  
}
```

}

Testing

Required

Unit Tests

Integration Tests

API Tests

Minimum coverage

70%

Database

Proper

Indexes

Foreign Keys

Constraints

Soft Delete

Audit Fields

Background Jobs

Use workers for

Resume Parsing

Email

AI Analysis

Report Generation

Storage

Support

Local

S3 Compatible Storage

DevOps

Provide

Dockerfile

Docker Compose

Environment Variables

Health Checks

README

One-command startup

`docker compose up`

CI/CD

GitHub Actions

Run

- Tests
- Lint
- Build Docker Image

AI Requirements

Avoid “LLM wrapper” projects.

Expected engineering includes:

- Embeddings
- Semantic Search

- RAG
- Prompt Versioning
- AI Caching
- Retry Logic
- Token Usage Tracking
- Cost Tracking

Bonus Features

- Voice Interview Agent
- WhatsApp Notifications
- OCR for Certificates
- Duplicate Candidate Detection
- AI Job Description Generator
- Candidate Skill Graph
- Referral System
- Resume Similarity Search
- AI Fraud Detection
- Interview Recording Analysis
- Multi-language Support
- Live Notifications
- Feature Flags
- Multi-region Deployment

Documentation

Include

Architecture Diagram

ER Diagram

API Documentation

Deployment Guide

Scaling Strategy

Explain how your system would support

- 10,000 companies
- 1 million candidates
- 500 recruiters simultaneously

Deliverables

GitHub Repository

Live Demo

Docker Setup

README

Architecture Diagram

API Documentation

ER Diagram

Short Demo Video (5–10 minutes)

Evaluation Rubric (100 Marks)

Area	Marks
System Architecture	15
Backend Engineering	20
AI Engineering	20
Frontend	10
Database Design	10
Security	10

DevOps & Docker	5
Testing	5
Documentation	5

What We Will Look For

We are **not** evaluating how many AI APIs you can integrate.

We are evaluating whether you can build software that could be deployed for real customers, maintained by a team, and scaled to thousands of organizations. We expect clean architecture, production-quality code, robust APIs, secure authentication, thoughtful AI integration with explainable outputs, comprehensive testing, and a deployment process that another engineer can reproduce with minimal effort.

Below is a production-grade open-source stack that is sufficient to build the entire assignment without relying heavily on proprietary services.

Category	Recommended	Alternatives
Backend	FastAPI	Django Ninja
Frontend	Next.js + React + Tailwind	Remix
Database	PostgreSQL	MariaDB
ORM	SQLAlchemy 2	Tortoise ORM
Migrations	Alembic	-
Cache	Redis	Valkey
Background Jobs	Celery + Redis	Dramatiq, ARQ

Authentication	FastAPI Users	Keycloak, Authentik
Object Storage	MinIO	SeaweedFS
Reverse Proxy	Traefik	Nginx
Containerization	Docker	Podman
Monitoring	Prometheus + Grafana	Netdata
Logging	Loki + Promtail	ELK/OpenSearch
API Testing	Pytest + HTTPX	Tavern
Load Testing	Locust	k6
CI/CD	GitHub Actions	Gitea Actions

AI & LLM

Frameworks

- LangGraph
- LlamaIndex
- Haystack
- PydanticAI

Haystack and LlamaIndex are mature open-source frameworks for production RAG pipelines, while LangGraph excels at agentic workflows.

Embeddings

Use open-source embedding models.

- BAAI/bge-large-en-v1.5
- BAAI/bge-m3
- Nomic Embed
- E5 Large

- Jina Embeddings

Local LLM

- Ollama
- vLLM
- llama.cpp

Models

- Qwen 3
- Llama 3.3
- Mistral
- Gemma

Vector Database

Recommended

- Qdrant ★★★★★

Alternatives

- Weaviate
- Milvus
- Chroma
- pgvector (PostgreSQL)

Resume Parsing

Instead of writing a parser from scratch, combine multiple open-source tools.

Document Parsing

- Docling ★★★★★
- Open Parse
- Markdown (Microsoft)
- Unstructured

These tools preserve layout, tables, headings, and document structure for downstream AI processing.

OCR

- Tesseract
- EasyOCR
- PaddleOCR

Resume Information Extraction

- spaCy
- HuggingFace Transformers
- sentence-transformers

Community recommendations for self-hosted resume parsing frequently combine Docling with spaCy or custom NLP pipelines.

Candidate Matching

Libraries

- sentence-transformers
- scikit-learn
- CrossEncoder
- HuggingFace Transformers

Matching Pipeline

Resume

↓

Extract Skills

↓

Embedding

↓

Vector Search

↓

Cross Encoder Re-ranking

↓

LLM Explanation

RAG

Recommended stack

Docling

↓

Chunking

↓

Qdrant

↓

Hybrid Search

↓

Cross Encoder

↓

LLM

Modern open-source RAG stacks increasingly use modular document parsing, hybrid retrieval, and reranking for higher-quality results.

Search

- OpenSearch
- Elasticsearch
- Meilisearch

Workflow Automation

- n8n
- Temporal
- Apache Airflow

Examples

- Interview reminder
- Resume uploaded
- Candidate shortlisted
- Email automation

Email

- Postal
- Listmonk
- Mailpit (development)

Notifications

- Novu ★★★★★

Supports

- Email
- SMS
- Push
- In-App notifications

Scheduling

- [Cal.com](https://cal.com)
- FullCalendar

PDF Generation

- WeasyPrint

- ReportLab

Charts

- Apache ECharts
- Recharts
- Chart.js

File Uploads

- MinIO
- Uppy
- FilePond

API Documentation

- OpenAPI
- Swagger UI
- ReDoc

RBAC

- Casbin ★★★★★

Supports

- Role-based access
- Organization policies
- Multi-tenant permissions

Security

- Passlib
- python-jose
- SlowAPI (rate limiting)
- Bandit
- Trivy

- OWASP Dependency Check

Observability

Monitoring

- Prometheus
- Grafana

Logging

- Loki
- Promtail

Tracing

- OpenTelemetry
- Jaeger

Feature Flags

- Flagsmith
- Unleash

Analytics

- Metabase
- Apache Superset

Testing

Backend

- Pytest
- Factory Boy
- Faker
- HTTPX

Frontend

- Vitest
- Playwright
- Cypress

DevOps

- Docker
- Docker Compose
- Kubernetes
- Helm
- ArgoCD
- GitHub Actions

Nice-to-have AI Libraries

- Instructor (structured LLM outputs)
- Guardrails AI (output validation)
- LiteLLM (multi-provider abstraction)
- Marvin
- DSPy (prompt optimization)

Suggested Production Architecture

Next.js Frontend

|

Traefik / Nginx

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FastAPI

|

PostgreSQL

Redis

Qdrant

MinIO

Celery Workers

Resume Parser

AI Matching

Email Service

Report Generator

LangGraph

LlamaIndex

Sentence Transformers

Ollama / vLLM

Prometheus

Grafana

Loki

Jaeger

This stack is entirely based on mature open-source technologies, giving the candidate exposure to the kinds of tools commonly used in production AI SaaS platforms while minimizing dependence on proprietary cloud services.